HTML (Type + Tab/Enter)

```
!              → Full HTML5 template
form           → <form></form>
input:text     → <input type="text">
input:number   → <input type="number">
input:password → <input type="password">
button         → <button></button>
div            → <div></div>
script:src     → <script src=""></script>
```

HTML Emmet Shortcuts

```
h2, label          → <h2></h2> ; <label></label>
input:text+button  → Input + Button combo
form>input:number*3 → Form with 3 number inputs
```

PHP (Type "php" + Tab)

```
php   → <?php ?>
fore  → foreach loop
if    → if statement
fun   → function template
```

Common Form

```
<form method="post" action="process.php">
  <input type="number" name="input1" id="input1" required>
  <button type="submit">Submit</button>
</form>
<div id="output"></div>
```

PHP Patterns (Universal Units)

```
function calc($ppl,$per,$cap,$uPrice=0,$sPrice=0){
  $need = $ppl*$per;
  $units = ceil($need/$cap);
  $left = $units*$cap-$need;
  $waste = $uPrice?($left/$cap)*$uPrice:$left*$sPrice;
  return [$units,$left,$waste]; }
// Pizza: calc(10,3,8,1050,0); // Venue:
calc(250,1,80,0,120); // Screen: calc(150,1,60,0,500);
```

Avg Speed (Q242)

```
function calc($tMin,$sUp,$sDn){
  $tUp=$tMin/60; $dist=$sUp*$tUp;
  return round(2*$dist/($tUp+$dist/$sDn));
}
```

Password Strength (Q242)

```
function str($p){
  $l=strlen($p);
  $u=preg_match('/[A-Z]/',$p);
  $w=preg_match('/[a-z]/',$p);
  $d=preg_match('/[0-9]/',$p);
  $s=preg_match('/[^A-Za-z0-9]/',$p);
  $e=str_ends_with($p,'*');
  if($l>=10&&$u&&$w&&$d&&$s&&$e) return
"Strong";
  if($l>=8&&$u&&$w&&$d) return "Medium";
  if($l>=6&&$u) return "Poor";
  return "Very Poor";
}
```

Cryptic Sequence (Q241)

```
function gen($seed,$n){
  $seq=[$seed]; $cur=$seed;
  for($i=1;$i<$n;$i++){
    $prod=1; $tmp=$cur;
    while($tmp>0){ $prod*=$tmp%10; $tmp=
floor($tmp/10); }
    $cur+=$prod; $seq[]=$cur;
  }
  return implode(", ",$seq);
}
```

PHP

```
<?php
$conn=new mysqli("localhost","root","","dbname");
$conn->query("CREATE DATABASE IF NOT EXISTS dbname");
$conn->select_db("dbname");
$conn->query("CREATE TABLE IF NOT EXISTS tbl(
  ID INT PRIMARY KEY,
  Name VARCHAR(100),
  Value INT
)");
?>
```

JAVASCRIPT PATTERNS

(Top of Every JS File)

```
const $ = id => document.getElementById(id);
const num = id => Number($(id).value);
const str = id => $(id).value;
const show = msg => $("output").innerHTML = msg;
const rand = (a,b) => Math.floor(Math.random()*(b-a+1))+a;
const dis = id => $(id).disabled = true;
```

SQL Quick Copy

```
-- Count by category
SELECT Grade, COUNT(*) AS total
FROM tbl GROUP BY Grade;

-- Sum revenue
SELECT Cat, SUM(Rev) AS tot
FROM tbl GROUP BY Cat;

-- Update low values
UPDATE tbl SET Grade='C'
WHERE Val<75 AND Grade<>'D';

-- Add bonus with cap
UPDATE tbl SET Val=Val+5
WHERE Val>80 AND Val+5<=90;
```

PHP-MYSQL: Connection + Setup

Insert Data

```
$conn->query("DELETE FROM tbl");
$conn->query("INSERT INTO tbl VALUES
  (1,'Item1',100),
  (2,'Item2',200)");
```

Select+Display

```
$res=$conn->query("SELECT * FROM tbl");
if($res->num_rows>0){
  while($r=$res->fetch_assoc()){
    echo $r["Name"]." - ".$r["Value"]."<br>";   }   }
```

Group by Count

```
$res=$conn->query("SELECT Grade, COUNT(*) AS cnt
  FROM tbl GROUP BY Grade");
while($r=$res->fetch_assoc())
  echo $r["Grade"].": ".$r["cnt"]."<br>";
```

Group by Sum

```
$res=$conn->query("SELECT Category, SUM(Revenue)
AS tot
  FROM tbl GROUP BY Category");
```

Update with Condition, Cap & Percentage

```
$conn->query("UPDATE tbl SET Grade='C'
  WHERE Value<75 AND Grade<>'D'");

$conn->query("UPDATE tbl SET Value=Value+5
  WHERE Value>80 AND Value+5<=90");

$conn->query("UPDATE tbl SET
Revenue=Revenue*1.10
  WHERE Revenue>70000");
```

Subquery (Top Seller)

```
$sql="SELECT Name, Revenue,
  CASE WHEN Revenue>(
    SELECT AVG(Revenue) FROM tbl t2
    WHERE t2.CatID=tbl.CatID
  ) THEN 'Top' ELSE 'Regular' END AS Lbl
  FROM tbl";
$res=$conn->query($sql);
```

Popularity Sort

```
$res=$conn->query("SELECT Course, COUNT(*) AS
cnt
  FROM tbl GROUP BY Course ORDER BY cnt
DESC");
```

Close Connection

```
$conn->close();
```

SQL Quick Copy

```
-- Percentage increase
UPDATE tbl SET Rev=Rev*1.10
WHERE Rev>70000;

-- Top N
SELECT * FROM tbl
ORDER BY Rev DESC LIMIT 2;

-- Having clause
SELECT ID, AVG(Qty) AS avg
FROM tbl GROUP BY ID
HAVING COUNT(DISTINCT date)>=2;
```

QUESTION IDENTIFIER

Question Says	Type	Section
pizza/venue/screen/minimum units/leftover/wasted	PHP-A	1
Average speed/uphill/downhill	PHP-B	2
Password strength/criteria/Strong/Medium/Poor	PHP-C	3
Cryptic sequence/seed/product of digits	PHP-D	4
electricity bill/rates/location/area/surcharge	PHP-E	5
database/GROUP BY/UPDATE/count/average	SQL	6

SECTION 1: PHP-A UNITS CALCULATOR

(Pizza/Venue/Screen - MOST COMMON!)

index.html:

```
<!DOCTYPE html>
<body> <h2>Calculator</h2>
<form method="post" action="process.php">
  //Pick Type A or B for wasted money formula
  <label>Input A:</label><input type="number" name="a" required><br>
  <label>Input B:</label><input type="number" name="b" required><br>
  <label>Input C:</label><input type="number" name="c" required><br>
  <button type="submit">Calculate</button>
</form>   </body>   </html>
```

process.php:

```
<?php
if ($_SERVER["REQUEST_METHOD"] === "POST") {
    $people = (int)$_POST["a"];      // students/attendees/people
    $perPerson = (int)$_POST["b"];   // slices per student / 1 for venues
    $capacity = (int)$_POST["c"];    // slices per pizza / seats per venue
        // Step 1: Total needed
    $totalNeeded = $people * $perPerson;
        // Step 2: Minimum units (ALWAYS ceil - can't book partial!)
    $totalUnits = (int)ceil($totalNeeded / $capacity);
        // Step 3: Leftover/Empty
    $leftover = ($totalUnits * $capacity) - $totalNeeded;
        // Step 4: Wasted Money - PICK ONE:
        // TYPE A: Fixed unit price (pizza 1050, venue 15000)
    $pricePerUnit = 1050;
    $wastedMoney = ($leftover / $capacity) * $pricePerUnit;
    // TYPE B: Per-seat/person price (ticket 500, cost 120)
    // $pricePerSeat = 500;  // $wastedMoney = $leftover * $pricePerSeat;
    echo "<h2>Results</h2>";
    echo "Total Units: " . $totalUnits . "<br>";
    echo "Leftover: " . $leftover . "<br>";
    echo "Wasted Money (BDT): " . number_format($wastedMoney, 2) . "<br>";  }?>
```

SECTION 2: PHP-B AVERAGE SPEED

process.php:

```
<?php
if ($_SERVER["REQUEST_METHOD"] === "POST") {
    $timeUpMinutes = (int)$_POST["timeUp"];
    $speedUphill = (int)$_POST["speedUp"];
    $speedDownhill = (int)$_POST["speedDown"];
        // Convert minutes to hours
    $timeUpHours = $timeUpMinutes / 60;
        // Calculate distance going up (speed × time)
    $distanceUp = $speedUphill * $timeUpHours;
        // Time to come down (same distance, different speed)
    $timeDownHours = $distanceUp / $speedDownhill;
        // Total distance and time
    $totalDistance = 2 * $distanceUp;
    $totalTime = $timeUpHours + $timeDownHours;
        // Average speed (return as integer)
    $avgSpeed = (int)round($totalDistance / $totalTime);
    echo "Average Speed: " . $avgSpeed . " km/h";    }   ?>
```

SECTION 3: PHP-C PASSWORD STRENGTH

process.php:

```
<?php
if ($_SERVER["REQUEST_METHOD"] === "POST") {
    $password = $_POST["password"];
    $length = strlen($password);
    $hasUpper = preg_match('/[A-Z]/', $password);
    $hasLower = preg_match('/[a-z]/', $password);
    $hasDigit = preg_match('/[0-9]/', $password);
    $hasSpecial = preg_match('/[^A-Za-z0-9]/', $password);
    $endsStar = str_ends_with($password, '*');
     // Strong, Medium & Poor; All types, ends with *
    if ($length >= 10 && $hasUpper && $hasLower && $hasDigit &&
$hasSpecial && $endsStar)
      { $strength = "Strong";  }
    elseif ($length >= 8 && $hasUpper && $hasLower && $hasDigit) {
$strength = "Medium";  }
    elseif ($length >= 6 && $hasUpper)
    { $strength = "Poor";  }
    else { $strength = "Very Poor"; }
    echo "Password Strength: " . $strength;    }   ?>
```

SECTION 4: PHP-D CRYPTIC SEQUENCE

process.php:

```
<?php
if ($_SERVER["REQUEST_METHOD"] === "POST") {
    $seed = (int)$_POST["seed"];
    $terms = (int)$_POST["terms"];
    $sequence = [$seed];
    $current = $seed;
    for ($i = 1; $i < $terms; $i++) {
        $product = 1;  //To calculate product of digits
        $temp = $current;
        while ($temp > 0)
      {   $digit = $temp % 10;
          $product *= $digit;
          $temp = (int)($temp / 10);  }
        // Add product to current
        $current = $current + $product;
        $sequence[] = $current;    }
    echo "Sequence: " . implode(", ", $sequence);   }   ?>
```

SECTION 5: PHP-E ELECTRICITY BILL ★★

process.php:

```
<?php
if ($_SERVER["REQUEST_METHOD"] === "POST") {
    $location = $_POST["location"]; // Dhaka, Chattogram, Other
    $area = $_POST["area"];            // Urban, Rural
    $units = (int)$_POST["units"];
    // Rate table
    $rates = [   "Dhaka" => ["Urban" => 7.5, "Rural" => 6.5],
        "Chattogram" => ["Urban" => 7.0, "Rural" => 6.0],
        "Other" => ["Urban" => 6.5, "Rural" => 5.5]   ];
    $rate = $rates[$location][$area];  // To Get Rate
    $baseBill = $units * $rate;  // To Calculate Base Bill
    $surcharge = 0;            // To Apply Surcharge
    if ($baseBill >= 500 && $baseBill <= 1000) {
        $surcharge = $baseBill * 0.05; }
    elseif ($baseBill > 1000)   {$surcharge = $baseBill * 0.10;  }
    $totalBill = $baseBill + $surcharge;
    echo "Total Bill: " . number_format($totalBill, 2) . " BDT";  }?>
```

SECTION 6.1: SQL (MySQL Database)

complete.php

```
<?php
// CONNECTION
$servername = "localhost";
$username = "root";
$password = "";
$dbname = "database_name";  // CHANGE THIS

$conn = new mysqli($servername, $username, $password);
if ($conn->connect_error) {  die("Connection failed: " . $conn-
>connect_error); }
// CREATE DATABASE
$conn->query("CREATE DATABASE IF NOT EXISTS
$dbname");
$conn->select_db($dbname);
// CREATE TABLE
$sql = "CREATE TABLE IF NOT EXISTS table_name (
    ID INT PRIMARY KEY,
    Name VARCHAR(100),
    CategoryID INT,
    CategoryName VARCHAR(100),
    Value INT,
    Grade CHAR(1)   )";
$conn->query($sql);
// INSERT DATA
$conn->query("DELETE FROM table_name");
$sql = "INSERT INTO table_name VALUES
    (1, 'Item1', 101, 'Category1', 85, 'B'),
    (2, 'Item2', 101, 'Category1', 92, 'A'),
$conn->query($sql);
echo "<h2>Database Setup Complete</h2><br>";
// TASK 1: COUNT BY CATEGORY/GRADE
echo "<h3>Task 1: Count by Grade</h3>";
$sql = "SELECT Grade, COUNT(*) AS total
        FROM table_name
        GROUP BY Grade
        ORDER BY Grade";
$result = $conn->query($sql);
if ($result->num_rows > 0) {
    echo "<table border='1'>";
    echo "<tr><th>Grade</th><th>Total</th></tr>";
    while($row = $result->fetch_assoc()){ echo "<tr><td>" .
$row["Grade"] . "</td>";
        echo "<td>" . $row["total"] . "</td></tr>";   }
    echo "</table><br>";  }
```

<<< Continued From Top Right Corner) >>>

SECTION 6.2: SQL (MySQL Database)

```
  .....
// TASK 2: UPDATE IF BELOW THRESHOLD
echo "<h3>Task 2: Update Low Values</h3>";
$sql = "UPDATE table_name
 SET Grade = 'C'
 WHERE Value < 75 AND Grade <> 'D'";
if ($conn->query($sql) === TRUE) {
 echo "Updated successfully<br><br>";  }
// TASK 3: ADD BONUS WITH CAP
echo "<h3>Task 3: Add Bonus</h3>";
$sql = "UPDATE table_name
 SET Value = Value + 5
 WHERE Value > 80 AND (Value + 5) <= 90";
if ($conn->query($sql) === TRUE) {
 echo "Bonus added successfully<br><br>";  }
// TASK 4: GROUP BY WITH ORDER (POPULARITY)
echo "<h3>Task 4: Popularity Ranking</h3>";
$sql = "SELECT CategoryName, COUNT(*) AS total
 FROM table_name
 GROUP BY CategoryName
 ORDER BY total DESC";
$result = $conn->query($sql);
if ($result->num_rows > 0) {
 echo "<table border='1'>";
 echo "<tr><th>Category</th><th>Count</th></tr>";
 while($row = $result->fetch_assoc()) {
 echo "<tr><td>" . $row["CategoryName"] . "</td>";
 echo "<td>" . $row["total"] . "</td></tr>";   }
 echo "</table><br>";  }
// BONUS: SUM REVENUE BY CATEGORY
$sql = "SELECT CategoryName, SUM(Revenue) AS total_revenue
 FROM table_name
 GROUP BY CategoryName";
// BONUS: UPDATE PERCENTAGE (10% BONUS)
$sql = "UPDATE table_name
 SET Revenue = Revenue * 1.10
 WHERE Revenue > 70000";
// BONUS: TOP SELLER LABEL (Compare to Category Average)
$sql = "SELECT Name, Value,
 CASE
 WHEN Value > (
 SELECT AVG(Value)
 FROM table_name t2
 WHERE t2.CategoryID = table_name.CategoryID
 ) THEN 'Top Seller'
 ELSE 'Regular Seller'
 END AS Label
 FROM table_name";
// DISPLAY FINAL TABLE
echo "<h3>Final Records</h3>";
$sql = "SELECT * FROM table_name ORDER BY ID";
$result = $conn->query($sql);
if ($result->num_rows > 0) {
 echo "<table border='1'>";
 echo "<tr><th>ID</th><th>Name</th><th>Value</th><th>Grade</
____th></tr>";
 while($row = $result->fetch_assoc()) {   echo "<tr>";
 echo "<td>" . $row["ID"] . "</td>";
 echo "<td>" . $row["Name"] . "</td>";
 echo "<td>" . $row["Value"] . "</td>";
 echo "<td>" . $row["Grade"] . "</td>";  echo "</tr>";   }
 echo "</table>";  }
$conn->close();   ?>
```

CRITICAL CHECKLIST

- ✅ **Always use ceil() for minimum units**
- ✅ **Check: if($_SERVER["REQUEST_METHOD"]==="POST")**
- ✅ **Check: if($result->num_rows>0) before while**
- ✅ **SQL: Use <> not !=**
- ✅ **Cast to int: (int)$_POST["field"]**

VS CODE SHORTCUTS (Type in .html)

- **! → Full HTML5 template**
- **form → Form tag**
- **input:number → Number input**
- **button → Button tag**

FORMULAS

Average Speed = (2 × Distance) / (TimeUp + TimeDown)
Percentage Bonus = Value × 1.10 (for 10%)
Surcharge = Bill × 0.05 (for 5%)
Ceil = ceil($value)
Random = rand($min, $max)